SE 216 – SOFTWARE PROJECT MANAGEMENT
PROJECT RISKS DOCUMENT

**PROJECT NAME:** NeuroMatter: Universal Local AI Hub

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| LIKELIHOOD RANK | RISK DESCRIPTION |
|---|---|
| 1 | High probability that local AI models (YOLOv8n, Qwen2-VL) will exceed the 2-second threshold on edge hardware. |
| 2 | Potential difficulties in ensuring seamless communication between different brands of Matter-compatible devices. |
| 3 | Technical challenges in synchronizing Flutter UI with C++/Python control layers and SQLCipher |
| 4 | Risk of sensor data from InfluxDB consuming local storage and RAM faster than anticipated. |
| 5 | Probability of the AI failing to maintain 95% detection accuracy in complex home environments. |
| 6 | Risk of hardware failure during extended power outages, impacting offline monitoring. |
| 7 | Low probability of 256-bit AES encryption being compromised in a local-only environment. |
| 8 | Low probability of system failure when managing 20+ concurrent smart devices. |
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| IMPACT RANK | RISK DESCRIPTION |
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| **1** | If processing exceeds 2 seconds, the system fails to prevent resident sensory overload, its core mission. |
| **2** | Exposure of sensitive behavioral logs violates the project's primary promise of data sovereignty. |
| **3** | Low accuracy in hazard detection could lead to household accidents for neurodivergent residents. |
| **4** | Failure to bridge software layers results in system downtime, leaving users without safety support |
| **5** | Total power loss renders all safety-critical monitoring and response functions inactive. |
| **6** | Limited compatibility restricts the "Universal" aspect of the hub and the device ecosystem. |
| **7** | May cause slight delays in automation but does not compromise immediate safety-critical functions. |
| **8** | Performance drops in Grafana visualization without affecting the core AI logic or response. |
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| LIKELIHOOD RANK | IMPACT RANK | COMBINED RANK | RISK DESCRIPTION |
|---|---|---|---|
| **1** | 1 | 2 | High probability that local AI models will exceed the 2-second threshold; if processing exceeds this, the system fails to prevent resident sensory overload, its core mission. |
| **3** | 4 | 7 | Technical challenges in synchronizing Flutter UI with C++/Python layers; failure results in system downtime, leaving users without safety support |
| **2** | 6 | 8 | Potential difficulties in ensuring seamless communication between different brands; limited compatibility restricts the "Universal" aspect of the hub. |
| **5** | 3 | 8 | Probability of the AI failing to maintain 95% detection accuracy; low accuracy could lead to household accidents for neurodivergent residents. |
| **7** | 2 | 9 | Low probability of 256-bit AES encryption being compromised; however, exposure of sensitive logs violates the primary promise of data sovereignty. |

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| **6** | 5 | 11 | Risk of hardware failure during extended power outages; total power loss renders all safety-critical monitoring and response functions inactive. |
| **4** | 8 | 12 | Risk of sensor data consuming local storage and RAM faster than anticipated; may cause slight delays in automation but does not compromise immediate safety. |
| **8** | 7 | 15 | Low probability of system failure when managing 20+ concurrent devices; performance drops in Grafana visualization without affecting core AI logic. |
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